Experiment No. 01

PART A (TO BE REFERRED BY STUDENTS)

A.1 Aim:

Installation and setup of R Programming Language and RStudio IDE.

A.2 Prerequisite:

1. Basic knowledge of computer operations.
2. Familiarity with Windows/Linux operating system.
3. Internet connectivity for downloading software.
4. Administrator privileges to install software.

A.3 Outcome:

After successful completion of this experiment, students will be able to:

1. Install the R Programming Language.
2. Install and configure the RStudio Integrated Development Environment (IDE).
3. Execute simple R commands using the R Console.
4. Create, save, and run an R Script.
5. Understand the basic interface of RStudio.

A.4 Theory:

R Programming

R is an open-source programming language widely used for statistical computing, data analysis, data visualization, and machine learning. It provides numerous built-in statistical functions and supports thousands of packages for advanced analytics.

RStudio IDE

RStudio is an Integrated Development Environment (IDE) that simplifies R programming. It provides a user-friendly interface for writing, executing, debugging, and managing R programs.

Features of R

- Open-source and free
- Cross-platform (Windows, Linux, macOS)
- Extensive statistical functions

- Powerful graphical capabilities
- Large collection of packages
- Supports data science and machine learning

Features of RStudio

- Script Editor
- Console Window
- Environment and History
- Files, Plots, Packages, Help, and Viewer
- Package Management
- Project Management

Basic Components of RStudio

1. Source Editor
2. Console
3. Environment/History
4. Files, Plots, Packages, Help, Viewer

Algorithm

Installation of R

1. Open a web browser.
2. Visit **https://cran.r-project.org**.
3. Download the latest version of R suitable for the operating system.
4. Run the installer.
5. Follow the installation wizard.
6. Complete the installation.

Installation of RStudio

1. Open a web browser.
2. Visit **https://posit.co/download/rstudio-desktop/**.
3. Download the latest version of RStudio Desktop.
4. Run the installer.
5. Complete the installation.
6. Launch RStudio.

Verification

1. Open RStudio.
2. Verify that the R Console appears.
3. Type the following commands:

```
2 + 3
print("Welcome to R Programming")
```

4. Press **Enter**.
5. Verify the output.
6. Create a new R Script.
7. Save the script.
8. Execute the script using **Run**.

PART B (TO BE COMPLETED BY STUDENTS)

Roll No.: C60

Name: Avdhut Parish Madhavi

Class: SE-C

Batch: C2

Date of Experiment: _____

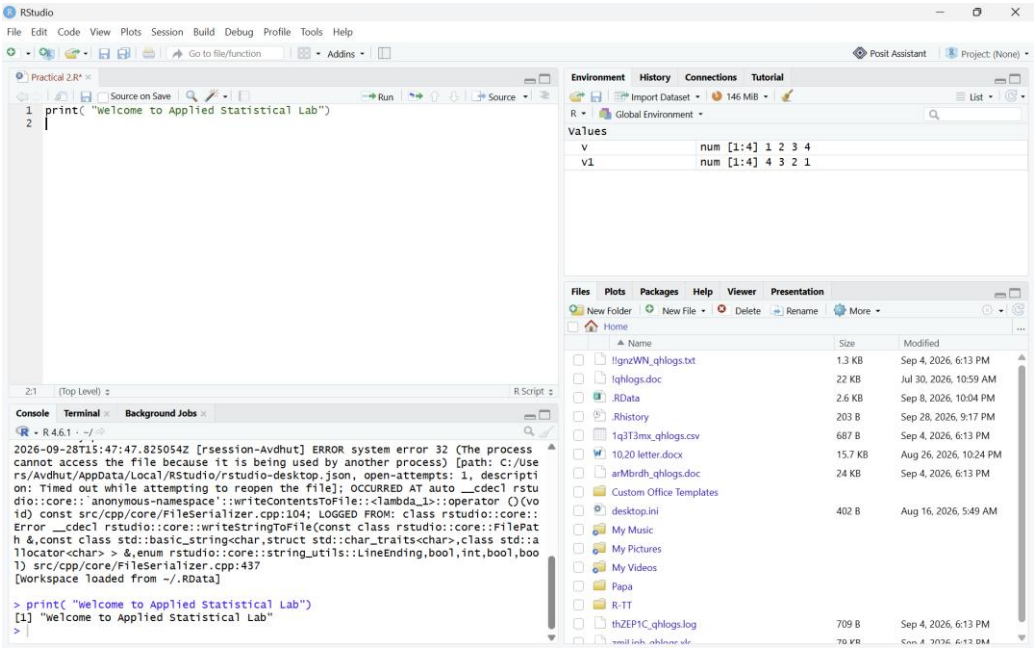
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B.1 Program Code:

```
print("Welcome to Applied Statistical Lab")
```

B.2 Output:



Expected Output:

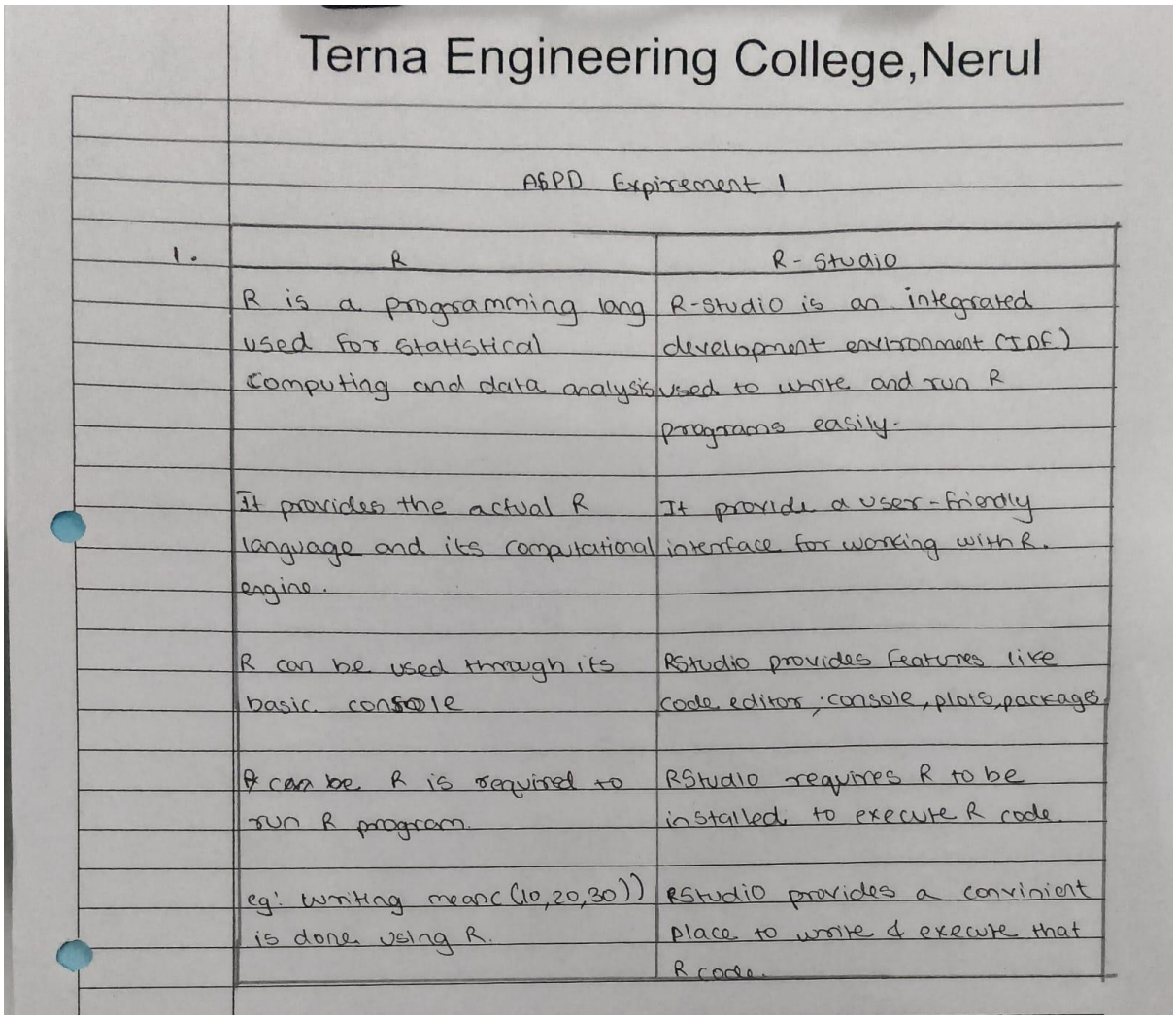
"Welcome to Applied Statistical Lab"

B.3 Conclusion:

The R Programming Language and RStudio IDE were successfully installed and configured. A basic R program was created, executed, and its output was verified successfully.

B.4 Questions of Curiosity:

Q.1 What is the difference between R and RStudio?

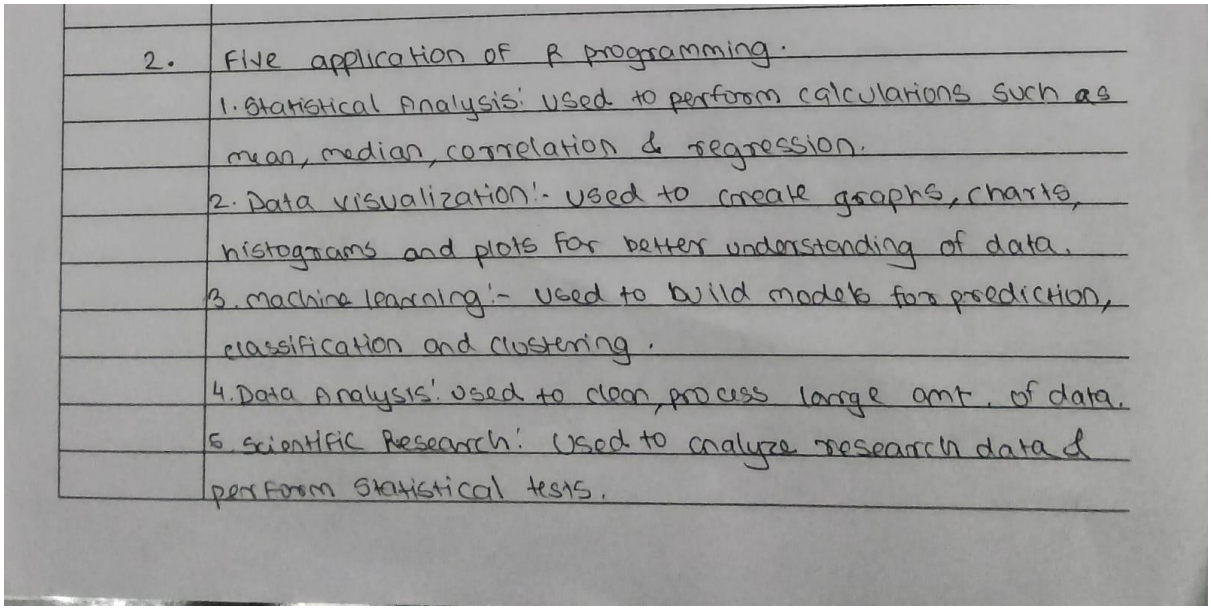


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ASPD Experiment 1

1.	R	R - Studio
	R is a programming lang used for statistical computing and data analysis	R-studio is an integrated development environment (IDE) used to write and run R programs easily.
	It provides the actual R language and its computational engine.	It provide a user-friendly interface for working with R.
	R can be used through its basic console	RStudio provides features like code editor, console, plots, packages
	R can be. R is required to run R program.	RStudio requires R to be installed to execute R code.
	eg: writing mean(c(10,20,30)) is done using R.	RStudio provides a convinient place to write & execute that R code.

Q.2 List any five applications of R Programming.

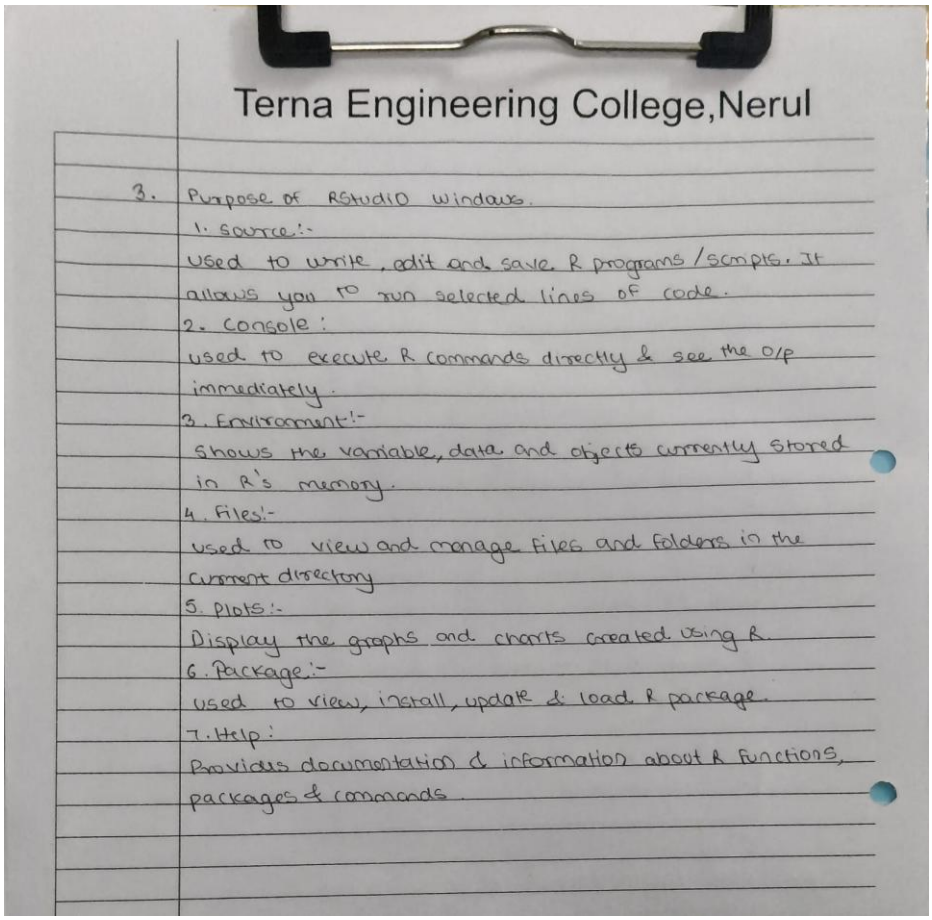


2. Five application of R programming.

1. Statistical Analysis: used to perform calculations such as mean, median, correlation & regression.
2. Data visualization:- used to create graphs, charts, histograms and plots for better understanding of data.
3. machine learning:- used to build models for prediction, classification and clustering.
4. Data Analysis: used to clean, process large amt. of data.
5. scientific Research: Used to analyze research data & perform statistical tests.

Q.3 Explain the purpose of the following RStudio windows:

- Source
- Console
- Environment
- Files/Plots/Packages/Help



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3. Purpose of RStudio windows.

1. Source:-
Used to write, edit and save R programs/scripts. It allows you to run selected lines of code.
2. Console:
used to execute R commands directly & see the o/p immediately.
3. Environment:-
Shows the variable, data and objects currently stored in R's memory.
4. Files:-
used to view and manage files and folders in the current directory
5. Plots:-
Display the graphs and charts created using R.
6. Package:-
used to view, install, update & load R package.
7. Help:
Provides documentation & information about R functions, packages & commands.